OPEN SOURCE · LEARNING ROADMAP

Python Fundamentals for AI and ML Roadmap

A curated, hands-on path from core Python programming to introductory machine learning — built around practical Jupyter notebooks for anyone learning by doing.

REPOSITORY

Jupyter Notebooks & Exercises

FORMAT

Guided Learning Roadmap

SECTION 01

Introduction

This repository brings together a set of hands-on Jupyter notebooks designed to guide a learner from the fundamentals of Python programming toward practical, real-world machine learning. Rather than presenting isolated snippets, each notebook builds on the last — forming a continuous, self-contained path that starts with variables and control flow and ends with model evaluation and neural networks.

The material is organized around three pillars: writing clean and correct Python code, working with and visualizing real data, and applying machine learning algorithms to solve problems of moderate complexity. Every concept is paired with runnable code, so the notebooks double as both a reference and a practice ground.

Whether you are picking up Python for the first time or looking to move from scripting into applied data science and machine learning, this roadmap is meant to be worked through sequentially, with each stage unlocking the tools needed for the next.

FOCUS	FORMAT	LEVEL
Python → Data Science → ML	**Jupyter Notebooks**	**Beginner to Intermediate**

SECTION 02

Learning Path

The roadmap is structured into progressive stages. Each stage groups related notebooks so you can move from foundational syntax to applied modeling in a logical sequence.

1 Beginner

- Python & Jupyter Notebook
- Variables
- Operators
- Data Types
- Control Flow

2 Core Python

- Lists
- Tuples
- Dictionaries
- Exception Handling
- File Handling
- Classes & Objects
- Standard Library

3 Data Analysis

- NumPy
- Pandas
- Data Wrangling
- Missing Values
- Duplicate Handling
- Outlier Detection
- Feature Encoding
- Data Visualization

4 Data Collection

- Web Scraping using BeautifulSoup

SECTION 02 · CONTINUED

Learning Path

5 Machine Learning Fundamentals

- Regression
- Classification
- Clustering (K-Means)
- DBSCAN

6 Model Evaluation

- Mean Squared Error
- Mean Absolute Error
- R^2 Score
- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score

7 Deep Learning Basics

- Introduction to TensorFlow
- Introduction to PyTorch
- Artificial Neural Networks

8 Practical Topics

- Model Deployment
- Model Integration
- Overfitting
- Underfitting
- Cross Validation

SECTION 03

Repository Usage

Every topic in this roadmap is accompanied by a dedicated Jupyter notebook that demonstrates the concept through working, practical code rather than theory alone. Notebooks are organized to mirror the stages above, so you can clone the repository and work through them in order, or jump directly to the topic you need.

Each notebook is self-contained: it includes explanations, runnable examples, and small exercises to reinforce the concept before moving to the next stage. This makes the repository equally useful as a first-time learning path or as a quick reference to revisit specific techniques.

SECTION 04

Skills Developed

Working through this roadmap builds a practical, end-to-end skill set spanning core programming and applied machine learning.

- ✓ **Python Programming**
- ✓ **Object-Oriented Programming**
- ✓ **File Processing**
- ✓ **Data Analysis**
- ✓ **Data Visualization**
- ✓ **Data Cleaning**
- ✓ **Machine Learning**
- ✓ **Model Evaluation**
- ✓ **Deep Learning Basics**

This roadmap is maintained as a living resource — notebooks are refined and extended over time as new techniques are added.